

# GANAMOORTHY SIVADHAS

IoT Developer · PCB Designer · Embedded Engineer

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## PROFESSIONAL SUMMARY

Electronics and Computer Science undergraduate (4th Year) with proven expertise in IoT system development, embedded firmware, and PCB design. Experienced in designing ESP32-based hardware-software solutions, integrating complex multi-sensor systems, and developing power-efficient firmware. Three-time national competition finalist with a track record of delivering innovative assistive and wearable technology solutions.

## EDUCATION

### Bachelor of Science (Hons) in Electronics and Computer Science 2023 – 2027

University of Kelaniya, Sri Lanka | 4th Year Undergraduate

**Relevant Coursework:** Embedded Systems, Circuit Design, IoT Development, Hardware Engineering, Signal Integrity, Digital Systems

**Activities:** Electronics Club Member, Tamil Association Member

## EXPERIENCE & PROJECTS

### Lumofit – Smart Posture Belt with AI Mar 2025 – May 2025

Hardware Engineer | IoT Developer · 🏆 Finalist – SLT Mobitel TechNovation 2025

- Spearheaded complete hardware architecture for a solar-powered assistive wearable featuring posture correction, health tracking, and emotion detection
- Integrated complex multi-sensor PCB (ECG, PPG, gyroscope, accelerometer, GPS, microphone) in a minimal form factor
- Developed power-efficient firmware for real-time bio signal processing and AI integration
- Delivered demonstrable impact in inclusive assistive technology for differently-abled individuals

### Smartens – Intelligent Wearable Assistance System Nov 2025 – Dec 2025

Hardware Engineer | IoT Developer · 🏆 Finalist – Pitch Arena, University of Ruhuna 2025

- Developed an ESP32-C3 based wearable for health monitoring, GPS tracking, and user assistance
- Integrated MAX30102 heart rate sensor, GPS module, OLED display, vibration motor, and audio output
- Implemented firmware for real-time heart rate detection, GPS parsing, BLE communication, and reminder alerts
- Designed a low-power embedded system with interactive UI and multi-function wearable features

### Guide Pro – Intelligent Presentation System Nov 2024 – Jan 2025

Embedded IoT Engineer

- Developed ESP32-powered edge computing device for real-time presentation assistance
- Designed custom PCB with INMP441 microphone and OLED display with minimal trace noise
- Integrated Vosk speech-to-text framework for intelligent offline speech analysis

Hardware Engineer | IoT Developer · 🏆 Finalist – Evolve Inter-University Challenge 2024

- Developed an **ESP32-based** IoT attendance system for real-time RFID-based student identification and tracking
- Integrated **MFRC522 RFID** module, WiFi connectivity, and LCD display with buzzer and LED indicators
- Designed reliable firmware with efficient data handling, duplicate-scan prevention, and real-time cloud synchronization

## TECHNICAL SKILLS

### Core Technical Skills

PCB Design

Embedded C/C++

IoT Systems

Python

MATLAB

### Programming & Web

C / C++ · Python · Java · JavaScript

HTML / CSS · MATLAB · Linux

### Tools & Platforms

EasyEDA · KiCAD · Altium · Proteus

Arduino IDE · VS Code · Git / GitHub

Kali Linux · Visual Studio

## NOTABLE ACHIEVEMENTS

### 🏆 Finalist – SLT Mobitel TechNovation 2025

Lumofit Smart Posture Belt — innovation in assistive technology for differently-abled individuals

### 🏆 Finalist – Pitch Arena 2025, Faculty of Engineering, University of Ruhuna

Smartens Smart Wearable Watch — real-time health monitoring and assistive wearable technology

### 🏆 Finalist – Evolve Inter-University Challenge 2024

RFID Attendance System — recognized for scalable IoT architecture and system design

## CERTIFICATIONS

PCB Design Mastery with Proteus — Alison | Aug 2025

Kali Linux Professional — Coursera | May 2024

Java & C Programming — SoloLearn

JavaScript Programming Certification — Cisco Networking Academy

## LANGUAGES

Tamil (Native) · English (Professional) · Sinhala (Conversational)